Contamination in mediated opinion-dynamics loops: linear baseline (per-node platform) and the gated identity

What this document is about. A platform (a model) is retrained each round on a population's opinions and feeds a prediction back to each person. Crucially, the platform reads each agent's *features* and emits a *per-node* prediction μ_i — different for different people — not a single broadcast value. We track one quantity, *contamination* s, the fraction of the population's opinion that is the platform's own recycled output, and establish two facts. **Piece 1 (linear).** Contamination has a closed form and is *value-blind*: it depends on the influence weights, not on what the platform predicts, so *retraining cannot change it*. Displacement and diversity, by contrast, *do* depend on the per-node predictions — and this is where "collapse" lives: the population collapses to a point only when the predictions are concentrated; spread (feature-driven) predictions preserve diversity. **Piece 2 (gated).** Adding a gate (people ignore predictions too far from their own view) turns s from a constant into a living variable with an exact growth law. All claims are certified by `theory_contamination_check.py`.

1 Notation

Symbol	Meaning
N	number of agents
$x(t) \in \mathbb{R}^N$	opinion vector at round t; component $x_i(t)$; initial $x(0)$
$\bar{x} := \frac{1}{N}\mathbf{1}^\top x$	population mean opinion
$\mathbf{1}$, I	all-ones vector, identity (size N)
$\lambda_i \in [0,1]$, $\Lambda := \mathrm{diag}(\lambda_i)$	agent i's self/innate weight (stubbornness)
W	social-influence matrix; $W_{ij} \geq 0$, row-stochastic ($W\mathbf{1} = \mathbf{1}$)
$C := (I - \Lambda)W$	composite social operator; $C\mathbf{1} = (I - \Lambda)\mathbf{1}$
$\boldsymbol{\mu} \in \mathbb{R}^N$	**per-node platform prediction**: μ_i is the model's output for agent i
$\nu\mathbf{1}$	a *uniform broadcast*: the special case $\mu_i \equiv \nu$ (the linear-literature baseline)
$f_i(t) \in [0,1]$	contamination tag: platform-share of $x_i(t)$; $s_t := \frac{1}{N}\sum_i f_i(t)$
$m \in [0,1)$	weight kept on the social signal; $1-m$ is the platform's injection weight
gated regime (Piece 2):	
$w \in [0,1]$	per-absorption blend weight toward μ_i
$\varepsilon > 0$, $G_t := \{i : \lvert\mu_i - x_i\rvert < \varepsilon\}$	confidence radius; contact set; $\mathrm{contact}_t := \lvert G_t\rvert/N$
$\langle g\rangle_{G_t} := \frac{1}{\lvert G_t\rvert}\sum_{i\in G_t} g_i$	average over the contact set

The experiments use $\boldsymbol{\mu}$, not $\nu\mathbf{1}$. Every MLP and LLM run serves a per-node prediction from features. The scalar broadcast $\nu\mathbf{1}$ appears only as the uniform special case (and in the prior linear literature); we carry it as a limiting case, not the model.

2 Standing assumptions

Assumption 1. W is row-stochastic ($W\mathbf{1} = \mathbf{1}$). *Everywhere.*

Assumption 2. $\lambda_i \equiv \lambda$. *Props. 2,4,5.*

Assumption 3. W doubly stochastic ($\mathbf{1}^\top W = \mathbf{1}^\top$); symmetric for Prop. 5(b). *Props. 4,5.*

Linear dynamics (per-node platform). Each agent updates to a blend of their innate opinion and a social average, where the social signal is the current opinions diluted, at weight $1 - m$, by the platform's per-node prediction μ_i:

$$x(t+1) = \Lambda x(0) + (I-\Lambda)W\big[m\,x(t) + (1-m)\boldsymbol{\mu}(t)\big] = \Lambda x(0) + mC\,x(t) + (1-m)(I-\Lambda)W\boldsymbol{\mu}(t). \quad (1)$$

Note the platform term keeps W: $(I-\Lambda)W\boldsymbol{\mu} \neq (I-\Lambda)\boldsymbol{\mu}$ because $\boldsymbol{\mu}$ is not constant. (W collapses out *only* for a uniform broadcast, $W(\nu\mathbf{1}) = \nu\mathbf{1}$.)

3 Part A — Contamination (value-blind)

The tag is a dye test (read this first). To measure "how much of an opinion came from the platform," run a second, parallel process on the *same* network: put dye of strength 1 on every platform source and 0 on every innate source, mix by the same weights each round, and read the dye concentration f_i at each agent. This is *not* the opinion equation with $\boldsymbol{\mu}$ replaced by 1; it is a separate bookkeeping flow. The opinion uses the real per-node values μ_i; the dye uses the label 1, which records only that the content is platform-origin — whatever its value. Consequently the dye injection is the uniform vector $\mathbf{1}$ (so $W\mathbf{1} = \mathbf{1}$ applies and W drops out), even though the opinion's platform term keeps W because $\boldsymbol{\mu}$ is not uniform. Everything in Part A is therefore identical whether the platform broadcasts one number or a different number to each person.

Idea. High level: the platform-share of an opinion depends on *how much* weight the platform carries and how it spreads, not on *what* the platform predicted. The dye test makes this exact: dye records provenance (1 = platform, 0 = innate), not value. Mathematically: the tag obeys a recursion whose injection is the constant $\mathbf{1}$ — because every platform output is dyed 1 regardless of its number.

Lemma 1 (tag dynamics and exactness)**.** *Let $f_i(t)$ be the platform-share of $x_i(t)$, $f(0) = 0$. Under Assumption 1,*

$$f(t+1) = mC\,f(t) + (1-m)(I-\Lambda)\mathbf{1}, \quad (2)$$

and $0 \le f_i(t) \le 1$ for all i, t (a genuine fraction).

Proof. Take the red (platform) fraction of each ingredient of $x_i(t+1)$ in (1), agent by agent:

$$f_i(t+1) = \underbrace{\lambda_i \cdot 0}_{\text{innate seed}} + \underbrace{(1-\lambda_i)\,m\textstyle\sum_j W_{ij}\,f_j(t)}_{\text{neighbor } j \text{ passes its own dye } f_j,\text{ not } x_j} + \underbrace{(1-\lambda_i)(1-m)\textstyle\sum_j W_{ij}\cdot 1}_{\text{platform content is 100\% red: dye 1, not } \mu_j}.$$

The innate seed is non-platform (dye 0); a neighbor contributes red in proportion to *its own* dye f_j, not its value x_j; the platform contributes dye 1, not its value μ_j (under terminal attribution every model output is 100% platform-origin, whatever its number). Since $\sum_j W_{ij} = 1$, the last sum is $(1-\lambda_i)(1-m)$. In vector form, using $C = (I-\Lambda)W$ and $W\mathbf{1} = \mathbf{1}$, this is (2). The injected dye vector is the constant $\mathbf{1}$ even though the injected *value* vector $\boldsymbol{\mu}$ is per-node (Part B uses the latter). Bounds: each $f_i(t+1)$ is a convex combination of 0, past dyes, and 1 (weights λ_i, $m(1-\lambda_i)$, $(1-m)(1-\lambda_i)$ sum to 1), so $f \in [0,1]$. □

Idea. High level: instead of simulating, get the long-run contamination in closed form. Mathematically: the tag recursion is a contraction plus a fixed injection, so it converges.

Proposition 1 (contamination fixed point)**.** *Under Assumption 1,* $f(t) \to B_\star := (1-m)(I - mC)^{-1}(I-\Lambda)\mathbf{1}$*.*

Proof. $f(t+1) = mCf(t) + b$, $b := (1-m)(I-\Lambda)\mathbf{1}$, is affine and a $\|\cdot\|_\infty$-contraction ($\|mC\|_\infty = m\max_i(1-\lambda_i) < 1$). Banach gives $B_\star = (I-mC)^{-1}b$. □

We write $s := B_\star$ for the per-agent contamination vector.

Idea. High level: with equal stubbornness the whole picture is one number, and you read off how stubbornness λ and platform strength $1-m$ trade off. Mathematically: $B_\star$ is a constant vector $s\mathbf{1}$.

Proposition 2 (scalar law)**.** *Under Assumptions 1–2,* $B_\star = s\mathbf{1}$ *with*

$$s = \frac{(1-m)(1-\lambda)}{\lambda + (1-\lambda)(1-m)} \qquad (\textit{platform pull / total pull}). \tag{3}$$

Proof. $C\mathbf{1} = (1-\lambda)\mathbf{1} \Rightarrow (I-mC)^{-1}\mathbf{1} = \mathbf{1}/(1-m(1-\lambda))$, and $(I-\Lambda)\mathbf{1} = (1-\lambda)\mathbf{1}$, so $B_\star = \frac{(1-m)(1-\lambda)}{1-m(1-\lambda)}\mathbf{1}$; the denominator is $\lambda + (1-\lambda)(1-m)$. □

Idea. High level: the punchline of the linear analysis. A platform that *retrains* — continually changing its per-node predictions in response to the population — changes *nothing* about contamination. Learning is inert here; only the gate (Part C) makes it matter. Mathematically: the tag recursion (2) contains no $\boldsymbol{\mu}$, so its limit s is the same for any prediction sequence.

Proposition 3 (degeneracy)**.** *Under terminal attribution (platform output is* 100% *platform content, whatever its value) and Assumption 1, let the platform retrain:* $\boldsymbol{\mu}(t) = G(x(t))$ *for any map* G (per-node, state-dependent)*. Then* $s = \lim_t s_t$ *equals its static value, although* $x(t)$ *depends on* $\{\boldsymbol{\mu}(t)\}$*.*

Proof. By Lemma 1 the tag recursion is autonomous — its injection $(1-m)(I-\Lambda)\mathbf{1}$ contains neither opinion values nor $\boldsymbol{\mu}$. Hence $\{f(t)\}$ and $B_\star$ are identical for every $\{\boldsymbol{\mu}(t)\}$ (scalar, per-node, static, or retrained): retraining moves *where* opinions sit, not *how much* platform mass accrues. □

4 Part B — Displacement and diversity (value-dependent)

Here the per-node predictions enter. For a static $\boldsymbol{\mu}$, the value fixed point of (1) is

$$x_\infty = A_\star x(0) + S_\star \boldsymbol{\mu}, \qquad A_\star := (I - mC)^{-1}\Lambda, \quad S_\star := (1-m)(I-mC)^{-1}(I-\Lambda)W, \tag{4}$$

and $S_\star \mathbf{1} = B_\star$, $A_\star \mathbf{1} + B_\star = \mathbf{1}$ (consistency with Part A: a uniform prediction $\boldsymbol{\mu} = \mathbf{1}$ recovers the tag).

Idea. High level: contamination is also the *displacement* weight — the population's average opinion is pulled a fraction s of the way to the platform's average prediction. Mathematically: the mean is the s-blend of the starting mean and the mean prediction $\bar{\mu}$.

Proposition 4 (equilibrium mean)**.** *Under Assumptions 1–3,* $\bar{x}_\infty = (1-s)\,\bar{x}_0 + s\,\bar{\mu}$*, where* $\bar{\mu} = \frac{1}{N}\mathbf{1}^\top \boldsymbol{\mu}$*.*

Proof. Left-multiply (4) by $\frac{1}{N}\mathbf{1}^\top$. With $\mathbf{1}^\top (I-mC)^{-1} = \mathbf{1}^\top/(1-m(1-\lambda))$ (Assumption 3) and $\mathbf{1}^\top (I-\Lambda)W = (1-\lambda)\mathbf{1}^\top$ (doubly stochastic), the innate coefficient is $\lambda/(1-m(1-\lambda)) = 1-s$ and the prediction coefficient is $(1-m)(1-\lambda)/(1-m(1-\lambda)) = s$. □

Idea. High level: *this is where "collapse" really lives, and where the scalar model was wrong.* A per-node platform does not squeeze everyone to one point — it *injects its own prediction-spread.* Diversity dies only if the predictions themselves are concentrated; feature-driven, spread-out predictions *preserve* diversity. Mathematically: variance has three terms, and the scalar claim $(1-s)^2$ is just the $\mathrm{Var}(\boldsymbol{\mu}) = 0$ special case.

Proposition 5 (equilibrium diversity)**.** *With* $\mathrm{Var}(z) := \frac{1}{N}\sum_i (z_i - \bar{z})^2$ *and Assumptions 1–2:* (a) *if* $W = I$*, then* $x_{\infty,i} = (1-s)x_{0,i} + s\mu_i$ *and*

$$\mathrm{Var}(x_\infty) = (1-s)^2\,\mathrm{Var}(x_0) + s^2\,\mathrm{Var}(\boldsymbol{\mu}) + 2s(1-s)\,\mathrm{Cov}(x_0, \boldsymbol{\mu}). \tag{5}$$

In particular: collapse $\mathrm{Var}(x_\infty) \to 0$ *requires* concentrated *predictions (*$\mathrm{Var}(\boldsymbol{\mu}) \to 0$*); a uniform broadcast (*$\mathrm{Var}(\boldsymbol{\mu}) = 0$*) recovers the scalar result* $\mathrm{Var}(x_\infty) = (1-s)^2\,\mathrm{Var}(x_0)$*.* (b) *for symmetric doubly-stochastic* W *the innate channel is additionally smoothed:* $(1-s)^2\,\mathrm{Var}(x_0)$ *is replaced by* $\frac{1}{N}\|A_\star \delta_0\|^2 \le (1-s)^2\,\mathrm{Var}(x_0)$*, while the platform injects* $\frac{1}{N}\|S_\star(\boldsymbol{\mu} - \bar{\mu}\mathbf{1})\|^2$*.*

Proof. (a) $W = I \Rightarrow A_\star = (1-s)I$, $S_\star = sI$, so $x_\infty = (1-s)x_0 + s\boldsymbol{\mu}$; variance of a sum gives (5). (b) From (4), $x_\infty - \bar{x}_\infty \mathbf{1} = A_\star \delta_0 + S_\star(\boldsymbol{\mu} - \bar{\mu}\mathbf{1})$ with $\delta_0 = x_0 - \bar{x}_0\mathbf{1}$ (using $A_\star\mathbf{1} = (1-s)\mathbf{1}$, $S_\star \mathbf{1} = s\mathbf{1}$). The first norm is bounded as in the uniform case (all non-Perron eigenvalues of $A_\star$ are $\le 1-s$); the second is the injected prediction-spread. □

Remark (the contraction is pinned — it does not reach zero here). Prop. 5 is a single equilibrium: uniform predictions contract diversity *once*, to $(1-s)^2\,\mathrm{Var}(x_0)$, and it *stays* there. It does not fall to zero, because the innate anchor $\Lambda x(0)$ in (1) re-injects the original opinions every round and holds the spread up. Diversity reaches zero only by either (i) starting from copies ($\mathrm{Var}(x_0) = 0$), or (ii) *removing the innate anchor* and predicting uniformly round after round — which is the bounded-confidence regime (no return-to-innate; Part C), not linear FJ. So the linear world is doubly safe: contamination s is fixed (Prop. 3) *and* diversity is floored at $(1-s)^2\,\mathrm{Var}(x_0)$ (this remark). The experiments drop both protections.

Reading of Part B. The per-node structure is the whole point: collapse (diversity death) is the special case where the model's predictions concentrate — which is exactly what an unanchored model does as it fits a narrowing population, and what an anchored (β-KL) model resists by keeping predictions feature-spread. The scalar model assumed $\mathrm{Var}(\boldsymbol{\mu}) = 0$ and so could only ever show collapse.

5 Part C — Gated regime: the contamination identity

What changes. The population is nonlinear. One round is a sweep of **peer** updates then a **gated absorption**, both using the per-node μ_i. *Peer:* an accepted pair ($|x_i - x_j| < \varepsilon$) meets in the middle, $x_i, x_j \leftarrow \frac{x_i+x_j}{2}$, $f_i, f_j \leftarrow \frac{f_i+f_j}{2}$. *Absorption:* agent $i \in G_t$ blends toward its own prediction, $x_i \leftarrow (1-w)x_i + w\mu_i$, $f_i \leftarrow (1-w)f_i + w \cdot 1$.

Idea. High level: first pin down where contamination comes from. People influencing each other never *creates* model-content; it only spreads it. So the platform is the sole source. Mathematically: the peer step preserves total tag mass $\sum f$.

Lemma 2 (peers conserve contamination)**.** *A peer midpoint preserves $f_i + f_j$; hence $\sum_i f_i$ is invariant under the peer sweep.*

Proof. $f_i' + f_j' = \frac{f_i+f_j}{2} + \frac{f_i+f_j}{2} = f_i + f_j$. □

Idea. High level: the central new result — the exact per-round growth of contamination, in measurable quantities: absorption strength w, how many are listening, and how clean they still are. Mathematically: Δs_t has a one-line closed form (and, like Part A, it does not read the values μ_i, only who is in contact).

Proposition 6 (exact s-identity)**.** *Over one round, with f taken before absorption,*

$$\Delta s_t = \frac{w}{N} \sum_{i \in G_t} (1 - f_i) = w \cdot \mathrm{contact}_t \cdot \langle 1 - f \rangle_{G_t}. \tag{6}$$

Proof. By Lemma 2 peers contribute nothing. Each $i \in G_t$ changes f_i by $[(1-w)f_i + w] - f_i = w(1 - f_i)$; agents $\notin G_t$ are unchanged. Sum, divide by N. □

Idea. High level: recovers the simple law we measured in simulations and says exactly when it holds. Mathematically: the special case where listeners are a representative sample.

Corollary 1 (empirical law)**.** *If $\langle f \rangle_{G_t} = s_t$, then $\Delta s_t = w \cdot \mathrm{contact}_t \cdot (1 - s_t)$.*

Idea. High level: explains the plateau — contamination can stall short of total when the model only reaches people it has already captured. Mathematically: the correction is a contamination–contact covariance that vanishes the increment.

Corollary 2 (detachment stall)**.** $\langle 1 - f \rangle_{G_t} = (1 - s_t) - [\langle f \rangle_{G_t} - s_t]$*. In detachment the gate stays open only for captured agents ($f_i \approx 1$ on G_t), so $\Delta s_t \approx 0$ and s stalls below* 1*.*

6 The hinge

Idea. High level: the one sentence that justifies the project. In the linear world s is a frozen constant retraining cannot touch (Prop. 3); the gate makes s a living, model-coupled variable (Prop. 6) with no closed form. Mathematically: the per-round tag map depends on the contact set, so it does not telescope into a fixed point.

Let P_t be the (mean-preserving) peer-averaging operator on tags and $D_t := \mathrm{diag}(w\,\mathbf{1}\{i \in G_t\})$. One round is $f_{t+1} = M_t f_t + c_t$ with $M_t = P_t(I - D_t)$, $c_t = P_t D_t \mathbf{1}$, both depending on G_t — hence on the opinions, hence on the model. Because M_t varies along the trajectory, $\prod_{\tau \le t} M_\tau$ does not telescope and no path-independent fixed point exists. Part D below carries out the minimal reduction that is still solvable; the variance floor and the endogenous-fraction collapse law (which feeds s_t into $\mathrm{Var}(\boldsymbol{\mu})$) build on it.

7 Part D — the gated phase boundary (two-body reduction)

Why a reduction. The hinge showed the full N-body gated loop has no closed-form fixed point. We reduce to the minimal model that still shows capture-vs-detachment: a single population position $y(t)$ and a platform anchored to a prior P_0. The platform fits the population but is held toward its prior, so its prediction is $\mu(t) = (1-\kappa)\,y(t) + \kappa\,P_0$, with anchor strength $\kappa \in (0,1)$ (increasing in the KL coefficient β). The gate is unchanged: the population absorbs only if $|\mu(t) - y(t)| < \varepsilon$, in which case $y(t+1) = (1-w)y(t) + w\mu(t)$; otherwise y is frozen.

Idea. High level: the full gated loop has no formula, but the question that matters — does the platform drag the population to its prior, or lose it? — does. It reduces to one number: the population's distance to the prior, against a threshold. Mathematically: in contact the dynamics are a one-dimensional linear contraction toward P_0, and the gate is a single inequality on that distance.

Proposition 7 (capture/detachment phase boundary)**.** *Let $d(t) := |y(t) - P_0|$. In contact, $y(t+1) - P_0 = (1 - w\kappa)(y(t) - P_0)$ and the gap is $|\mu(t) - y(t)| = \kappa\,d(t)$. Hence:*

- *if $d(0) < \varepsilon/\kappa$: contact persists, $d(t) = (1-w\kappa)^t d(0) \to 0$, and $y(t) \to P_0$* (capture)*;*
- *if $d(0) > \varepsilon/\kappa$: the gate never opens and $y(t) \equiv y(0)$* (detachment)*.*

The basin boundary is $d(0) = \varepsilon/\kappa$.

Proof. In contact, $y(t+1) - P_0 = (1-w)(y(t) - P_0) + w(\mu(t) - P_0)$ with $\mu(t) - P_0 = (1-\kappa)(y(t) - P_0)$, so $y(t+1) - P_0 = (1 - w\kappa)(y(t) - P_0)$; since $0 < w\kappa < 1$, $d(t)$ strictly decreases and the gap $\kappa d(t)$ with it. If $\kappa d(0) < \varepsilon$ the gap stays below ε for all t — contact is self-reinforcing, $d(t) \to 0$, $y \to P_0$. If $\kappa d(0) \ge \varepsilon$, y is frozen, $d(t) = d(0)$, and the gap stays $\ge \varepsilon$. Both regions are absorbing; the boundary is $d(0) = \varepsilon/\kappa$. □

Plain reading. Close enough to the prior to listen ⇒ being pulled toward it brings you closer ⇒ you keep listening: capture snowballs. Too far to listen ⇒ you never move ⇒ you stay too far: detachment is permanent.

Idea. High level: a counterintuitive measured fact — a *stronger* anchor makes the platform *lose* the population more easily — now has a one-line reason. Mathematically: the capture region ε/κ shrinks as κ grows.

Corollary 3 (anti-monotone in the anchor)**.** *The capture region $\{d(0) < \varepsilon/\kappa\}$ shrinks as κ increases. Writing $\kappa = aw/(1+aw)$ for an anchor weight aw, the boundary is $\varepsilon/\kappa = \varepsilon(1+aw)/aw$ — the gate condition observed empirically (08d).*

Idea. High level: a spread-out population does not all go one way — it *splits*. The agents close enough to the prior are captured; the rest are stranded where they started. This is the stranded minority, and the reason contamination stalls. Mathematically: each agent's fate is decided independently by its own distance to P_0.

Corollary 4 (dissociation and the contamination stall)**.** *For a population spread over $\{y_i\}$, agent i is captured iff $|y_i - P_0| < \varepsilon/\kappa$. The population dissociates into a captured mass at P_0 and a detached residue frozen at $\{y_i\}$. The detached fraction never absorbs, so its tag stays* 0*; by Cor. 2 the mean contamination s stalls below* 1 *at the captured fraction.*

Idea. High level: at the threshold the outcome is knife-edge, so tiny differences decide it — which is why repeated runs with different randomness land in different basins. Mathematically: the deterministic boundary is sharp, so finite-N fluctuations flip marginal agents across it.

Corollary 5 (bistability)**.** *Both basins are absorbing, so for $d(0)$ near ε/κ the outcome is set by which side of the threshold the realised initial condition falls on; finite-N noise yields seed-dependent splits.*

Scope. This is the reduction, not the full N-body loop: it assumes the anchored-fit prediction $\mu = (1-\kappa)y + \kappa P_0$ and treats agents as independent (no peer coupling). It captures capture, detachment, dissociation, and the boundary ε/κ, but not peer-driven cluster merging (neighbours pushing an agent across the threshold), which remains simulation-only. What is new and proven is the closed-form boundary ε/κ and that one mechanism explains the anti-monotone anchor effect (Cor. 3), the dissociation (Cor. 4), and the contamination stall.